

Ex. No. : 04

Date: 25/02/2026

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Gesture-Based Interaction App

Aim

To develop a gesture-based mobile application that detects double-tap and swipe gestures, displays text on double tap, navigates between screens on subsequent double taps, and shows toast messages on left and right swipe actions.

Procedure:

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <TextView
        android:id="@+id/gesture_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:visibility="gone"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Activity_second.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Second Screen"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.xml

```

package com.example.gestureapp

import android.content.Intent
import android.os.Bundle
import android.view.GestureDetector
import android.view.MotionEvent
import android.view.View
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.GestureDetectorCompat

class MainActivity : AppCompatActivity(), GestureDetector.OnGestureListener,
GestureDetector.OnDoubleTapListener {

    private lateinit var gestureDetector: GestureDetectorCompat
    private lateinit var gestureText: TextView
    private var doubleTapCount = 0

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
    }

```

```

        gestureText = findViewById(R.id.gesture_text)
        gestureDetector = GestureDetectorCompat(this, this)
        gestureDetector.setOnDoubleTapListener(this)
    }

    override fun onTouchEvent(event: MotionEvent): Boolean {
        gestureDetector.onTouchEvent(event)
        return super.onTouchEvent(event)
    }

    override fun onDown(e: MotionEvent): Boolean = true

    override fun onShowPress(e: MotionEvent) {}

    override fun onSingleTapUp(e: MotionEvent): Boolean = true

    override fun onScroll(e1: MotionEvent?, e2: MotionEvent, distanceX: Float,
        distanceY: Float): Boolean = true

    override fun onLongPress(e: MotionEvent) {}

    override fun onFling(e1: MotionEvent?, e2: MotionEvent, velocityX: Float,
        velocityY: Float): Boolean {
        if (e1 != null) {
            val diffX = e2.x - e1.x
            if (diffX > 0) {
                Toast.makeText(this, "Right Swipe", Toast.LENGTH_SHORT).show()
            } else {
                Toast.makeText(this, "Left Swipe", Toast.LENGTH_SHORT).show()
            }
        }
        return true
    }

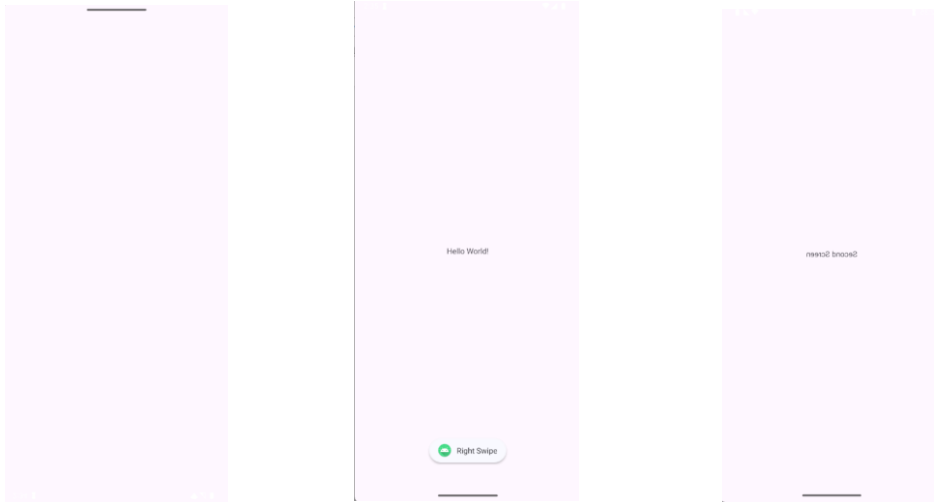
    override fun onSingleTapConfirmed(e: MotionEvent): Boolean = true

    override fun onDoubleTap(e: MotionEvent): Boolean {
        if (doubleTapCount == 0) {
            gestureText.visibility = View.VISIBLE
            doubleTapCount++
        } else {
            val intent = Intent(this, SecondActivity::class.java)
            startActivity(intent)
        }
    }

```

```
    }  
    return true  
}  
  
override fun onDoubleTapEvent(e: MotionEvent): Boolean = true  
}
```

Output



Result:

Hence the experiment is completed and the output is verified..